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|--------------------------------|-------------------|
| Course Name: signal and system | Date: 12/4/2022   |
| lecturer: Dr. Aws Al-Qaisi     | Exam Time: 40 min |

1-given  $x(t)=-3t+15$  where,  $4 \leq t \leq 5$ . find the time interval for  $x(-2t+4)$ , Select one:

- a.  $-0.8 \leq t \leq -0.4$    b.  $-1.75 \leq t \leq -1.5$    c.  $-0.75 \leq t \leq -0.5$    d.  $0.5 \leq t \leq 0.75$    e. None

2-given  $x(t)=t+4$  where  $-4 \leq t \leq -2$ , find  $x(-2t+4)$

- a.  $x(t)=-4t+4$  where,  $1 \leq t \leq 1.5$    b.  $x(t)=-4t+6$  where,  $1 \leq t \leq 1.5$   
c.  $x(t)=-4t+4$  where,  $1.5 \leq t \leq 2.5$    d.  $x(t)=-4t+6$  where,  $1.5 \leq t \leq 2.5$    e. None

3-if  $x(t)=r(t+3)-r(2t)$ , the  $x(t)$  during the time intervals  $-3 \leq t \leq 0$  can be written as

- a. 3   b.  $t+3$    c.  $3-t$    d.  $t-3$    e. None

4- $x(t)=r(t)-2r(t-2)$  find  $x(-2t+2)$ .

- a.  $x(-2t+2)=2-2t$  ( $-1 < t < 0$ ),  $x(-2t+2)=2t+2$  ( $0 < t < 1$ )  
b.  $x(-2t+2)=2-2t$  ( $0 < t < 1$ ),  $x(-2t+2)=2t+2$  ( $-1 < t < 0$ )  
c.  $x(-2t+2)=2t+2$  ( $0 < t < 1$ ),  $x(-2t+2)=2t+2$  ( $-2 < t < 0$ )  
d.  $x(-2t+2)=-2t+2$  ( $0 < t < 1$ ),  $x(-2t+2)=2t+2$  ( $-1 < t < 0$ )  
e. None

5-If  $x(n)=(0,0,1,2,3,4,0,0)$  then  $x(n-3)$  is

- a.  $(0,0,0,2,4,6,8,0,0)$    b.  $(0,0,0,1,2,3,4,0,0)$    c.  $(1,2,3,4,0,0,0,0,0)$    d.  $(0,0,0,0,1,2,3,4)$

6- The impulse response  $h(n)$  of a linear time invariant system is given by:

$$h(n) = u(n+3) + u(n-2) + 2u(n-7), \text{ where } u(n) \text{ is the unit step sequence.}$$

The above system is:?

- a. stable and causal   b. stable but not causal   c. causal but unstable  
d. unstable and not causal   e. None
- 

7- Determine the product of two signals:  $x_1(n) = \{2, 1, 1.5, 3\}$ ;  $x_2(n) = \{1, 1.5, 0, 2\}$ .

- a.  $\{2, 1.5, 0, 6\}$    b.  $\{2, 1.5, 6, 0\}$    c.  $\{2, 0, 1.5, 6\}$    d.  $\{2, 1.5, 0, 3\}$
- 

8-  $h_1(t)$  convolved with  $h_2(t) = h_2(t)$  convolved with  $h_1(t)$

- a. Associative property   b. Distributive property  
c. Commutative property   d. Multiplication property
- 

9- find  $y(n) = x(n) * h(n)$ , where  $x(n) = [1, 1]$  and  $h(n) = [1, 1]$

- a.  $[1, 2, 1]$    b.  $[1, 2, 3]$    c.  $[3, 1, 1]$    d.  $[1, 1, 2]$
- 

10- Find the fundamental period and frequency of the following signal

$$x(t) = 2\cos(6\pi t + \pi/6)$$

- a.  $T=0.16$  sec,  $f=6$  Hz   b.  $T=10$  sec,  $f=0.1$  Hz   c.  $T=0.33$  sec,  $f=3$  Hz   d.  $T=5$  sec,  $f=0.2$  Hz
- 

11- Is the following signal an energy signal

$$x(t) = u(t) - u(t-1) + t(t-4)$$

- a. YES   b. NO
- 

12- Find the value of  $2h[n] * \delta(n+1)$

- a.  $2h[n-2]$    b.  $2h[n]$    c.  $2h[n-1]$    d.  $2h[n+1]$



13-  $h_1(t) * h_2(t) * h_3(t) = h_3(t) * h_2(t) * h_1(t)$

this property of convolution process is called \_\_\_\_\_

- a. Associative property                      b. Commutative property  
c. Distributive property                      d. convolved with impulse property.

14- Noise generated by an amplifier of radio is an example for?

- a. Discrete signal      b. Deterministic signal      c. Random signal      d. Periodic signal

15- any time delay or time in advance in the input signal leads to a different time delay or time in advance in the output signal :

in this case the system is called

- a. Time invariant systems                      b. Memory less systems  
c. Causal systems                                  d. message frequency

| Question | 1. | 2. | 3. | 4. | 5. | 6. | 7. | 8. | 9. | 10 |
|----------|----|----|----|----|----|----|----|----|----|----|
| Answer   |    |    |    |    |    |    |    |    |    |    |
| Question | 11 | 12 | 13 | 14 | 15 |    |    |    |    |    |
| Answer   |    |    |    |    |    |    |    |    |    |    |

a. Time invariant systems

b. Memory less systems

c. Causal systems

d. message frequency

| Question | 1. | 2. | 3. | 4. | 5. | 6. | 7. | 8. | 9. | 10. |
|----------|----|----|----|----|----|----|----|----|----|-----|
| Answer   | A  | D  | C  | C  | D  | B  | A  | C  | A  | C   |
| Question | 11 | 12 | 13 | 14 | 15 |    |    |    |    |     |
| Answer   | A  | D  | B  | C  | A  |    |    |    |    |     |

29/3/20